Finite-Interval Compression and Transverse Control

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Abstract

We examine compression of the full-line projector relations for the occupied-band covariances of a one-dimensional Wilson–Dirac chain. The full-line squared-difference and transverse identities imply positive finite-interval inequalities with exact boundary-leakage remainders. A stronger transverse domination follows, together with an explicit noncommutation example showing why the corresponding full-line operator products cannot be transported unchanged to finite intervals.

Operator setting

Let P_m and P_0 be the full-line occupied-band projections for the Wilson–Dirac Bloch symbol

$$h_m(k) = \sin k\,\sigma_x + (1 + m - \cos k)\sigma_z, \quad P_m(k) = \frac{1}{2}\left(I - \frac{h_m(k)}{\sqrt{m^2 + 4(1+m)\sin^2(k/2)}}\right).$$

Fix an interval of L two-orbital cells, with projection χ, $N = 2L$, $\mu > 0$, and $m = \mu/L$. After a common interval-preserving rotation and staggering, the full-line covariance difference is $T = P_m - P_0 = S + D$, where

$$S = -\frac{g_r}{2}Q_0, \qquad D = \frac{a_r}{2}Q_\perp, \qquad G = \frac{g_r}{2}I,$$

$$r = \frac{\mu}{N+\mu}, \qquad u_r = \frac{|\sin(k/2)|}{\sqrt{\sin^2(k/2) + r^2\cos^2(k/2)}}, \quad g_r = 1 - u_r, \quad a_r = \frac{r\cos(k/2)}{\sqrt{\sin^2(k/2) + r^2\cos^2(k/2)}}.$$

The Hermitian involutions $Q_0, Q_\perp$ anticommute. In particular, the **full-line** identities are

$$T^2 = G, \qquad D^2 = G - G^2, \qquad 0 \le G \le \frac{1}{2}I.$$

Set $\Delta_N = \chi T\chi$, $D_N = \chi D\chi$, and $G_N = \chi G\chi$, as operators on $\operatorname{ran}\chi$.

Theorem: exact boundary remainders

For every positive integer L and $\mu > 0$,

$$\boxed{G_N - \Delta_N^2 = \chi T(I-\chi)T\chi \ge 0,}$$

$$\boxed{G_N - G_N^2 - D_N^2 = \chi G(I-\chi)G\chi + \chi D(I-\chi)D\chi \ge 0.}$$

Therefore,

$$\boxed{0 \le \Delta_N^2 \le G_N \le \tfrac{1}{2} I, \quad 0 \le D_N^2 \le G_N - G_N^2 \le G_N.}$$

Taking the finite trace in the first relation gives the exact leakage formula

$$\operatorname{Tr} G_N - \|\Delta_N\|_{\mathrm{HS}}^2 = \|(I-\chi)T\chi\|_{\mathrm{HS}}^2.$$

Proof. For any bounded self-adjoint X, insertion of $I = \chi + (I-\chi)$ gives

$$\chi X^2 \chi - (\chi X \chi)^2 = \chi X (I-\chi) X \chi \ge 0.$$

Apply this identity to T and use $T^2 = G$ for the first formula. For the second, apply it to both G and D, obtaining

$$G_N^2 = \chi G^2 \chi - \chi G (I-\chi) G \chi, \quad D_N^2 = \chi D^2 \chi - \chi D (I-\chi) D \chi.$$

Substitute the full-line equality $G - G^2 - D^2 = 0$. Every remainder is a positive square. Because $0 \le G_N \le I/2$, the remaining order comparisons follow. The trace identity is the Hilbert–Schmidt norm identity for $(I-\chi)T\chi$. □

Proposition: the compressed majorant need not commute with the compressed reference

Let $B_N = \chi P_0 \chi$ in the scalarized basis. At $N = 4$, define

$$c_1 = \frac{1}{\pi} \int_0^\pi g_r(k) \cos k \, dk.$$

The scalarized sine kernel B_4 and the same-parity Toeplitz matrix G_4 satisfy

$$\boxed{[G_4, B_4]_{0,1} = -\frac{2c_1}{3\pi} \ne 0.}$$

Indeed $g_r(k) = 1 - (1 + r^2 \cot^2(k/2))^{-1/2}$ decreases strictly over $(0, \pi)$, so integration by parts shows $c_1 = -\pi^{-1} \int_0^\pi g_r'(k) \sin k \, dk > 0$. The matrix formula follows by direct multiplication of the sine-kernel entries $B_{ij} = \sin[\pi(i-j)/2]/[\pi(i-j)]$ off the diagonal and the parity-supported Fourier coefficients of G_4. □

The full-line $[G, P_0] = 0$ is nevertheless sufficient for the Wilson staggered-pairing inequality: that proof decomposes the interval-supported vector using the full-line projector before evaluating the compressed expectation. The finite model does not require the false identity $[G_N, B_N] = 0$.

Scientific scope and acknowledgments

The theorem pertains to the specified infinite-chain band projections and finite interval compressions, for every $L \ge 1$ and $\mu > 0$. No reference-modular, Coulomb, all-degree nonlinear, or cross-boundary entropy limit follows solely from these identities. The leading relative-entropy law follows only after combining these identities with the separate bounded nonlinear, boundary-entropy and weighted modular estimates, as done in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law.*

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